CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 21: Trigonometrical Identities

Topic-wise short notes with solved examples

An identity is an equation that is true for every value of the angle. This chapter covers the basic identities, the ratios of complementary angles, and the methods for proving identities.

1 The ratios and their relations

- For an acute angle A of a right-angled triangle: $\sin A = \dfrac{\text{perpendicular}}{\text{hypotenuse}}$, $\cos A = \dfrac{\text{base}}{\text{hypotenuse}}$, $\tan A = \dfrac{\text{perpendicular}}{\text{base}}$.
- **Reciprocal relations**: $\operatorname{cosec} A = \dfrac{1}{\sin A}$, $\sec A = \dfrac{1}{\cos A}$, $\cot A = \dfrac{1}{\tan A}$.
- **Quotient relations**: $\tan A = \dfrac{\sin A}{\cos A}$ and $\cot A = \dfrac{\cos A}{\sin A}$.

2 The three basic identities

$$\sin^2 A + \cos^2 A = 1 \qquad 1 + \tan^2 A = \sec^2 A \qquad 1 + \cot^2 A = \operatorname{cosec}^2 A$$

Identity	Other useful forms
$\sin^2 A + \cos^2 A = 1$	$1 - \sin^2 A = \cos^2 A$; $1 - \cos^2 A = \sin^2 A$
$1 + \tan^2 A = \sec^2 A$	$\sec^2 A - \tan^2 A = 1$; $(\sec A + \tan A)(\sec A - \tan A) = 1$
$1 + \cot^2 A = \operatorname{cosec}^2 A$	$\operatorname{cosec}^2 A - \cot^2 A = 1$; $(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$

3 Proving identities

- Start with the more complicated side and simplify it until it becomes the other side.
- If no shorter way is visible, change everything to $\sin$ and $\cos$.
- For a sum of fractions, take the LCM.
- For $1 - \sin A$ or $1 - \cos A$ in a denominator, multiply the numerator and denominator by the conjugate $1 + \sin A$ or $1 + \cos A$.
- Use algebraic identities: $a^2 - b^2 = (a + b)(a - b)$ and $(a + b)^2 = a^2 + 2ab + b^2$.

Example 3.1 Prove that $\sin^4 A - \cos^4 A = 2\sin^2 A - 1$.

SOLUTION

$\text{LHS} = (\sin^2 A + \cos^2 A)(\sin^2 A - \cos^2 A) = \sin^2 A - \cos^2 A$

$= \sin^2 A - (1 - \sin^2 A)$

$= 2\sin^2 A - 1 = \text{RHS}$

Example 3.2 Prove that $\dfrac{\cos A}{1-\sin A}+\dfrac{\cos A}{1+\sin A}=2\sec A$.

SOLUTION

$$\text{LHS}=\frac{\cos A(1+\sin A)+\cos A(1-\sin A)}{(1-\sin A)(1+\sin A)}=\frac{2\cos A}{1-\sin^2 A}$$
$$=\frac{2\cos A}{\cos^2 A}=\frac{2}{\cos A}$$
$$=2\sec A=\text{RHS}$$

Example 3.3 Prove that $\sqrt{\dfrac{1+\sin A}{1-\sin A}}=\sec A+\tan A$.

SOLUTION

Multiply the numerator and denominator inside the root by $(1+\sin A)$:

$$\text{LHS}=\sqrt{\frac{(1+\sin A)^2}{1-\sin^2 A}}=\sqrt{\frac{(1+\sin A)^2}{\cos^2 A}}=\frac{1+\sin A}{\cos A}$$
$$=\frac{1}{\cos A}+\frac{\sin A}{\cos A}$$
$$=\sec A+\tan A=\text{RHS}$$

Example 3.4 Prove that $(\operatorname{cosec} A-\sin A)(\sec A-\cos A)(\tan A+\cot A)=1$.

SOLUTION

$$\operatorname{cosec} A-\sin A=\frac{1-\sin^2 A}{\sin A}=\frac{\cos^2 A}{\sin A}\text{ and }\sec A-\cos A=\frac{1-\cos^2 A}{\cos A}=\frac{\sin^2 A}{\cos A}$$
$$\tan A+\cot A=\frac{\sin^2 A+\cos^2 A}{\sin A\cos A}=\frac{1}{\sin A\cos A}$$
$$\text{LHS}=\frac{\cos^2 A}{\sin A}\times\frac{\sin^2 A}{\cos A}\times\frac{1}{\sin A\cos A}$$
$$=1=\text{RHS}$$

Example 3.5 If $\sin A+\cos A=p$ and $\sec A+\operatorname{cosec} A=q$, show that $q(p^2-1)=2p$.

SOLUTION

$$p^2-1=\sin^2 A+\cos^2 A+2\sin A\cos A-1=2\sin A\cos A$$
$$q=\frac{1}{\cos A}+\frac{1}{\sin A}=\frac{\sin A+\cos A}{\sin A\cos A}=\frac{p}{\sin A\cos A}$$
$$q(p^2-1)=\frac{p}{\sin A\cos A}\times 2\sin A\cos A=2p$$

4 Complementary angles

- Two angles are **complementary** if their sum is 90°.

sin, tan, sec	cos, cot, cosec
$\sin(90^\circ - A) = \cos A$	$\cos(90^\circ - A) = \sin A$
$\tan(90^\circ - A) = \cot A$	$\cot(90^\circ - A) = \tan A$
$\sec(90^\circ - A) = \operatorname{cosec} A$	$\operatorname{cosec}(90^\circ - A) = \sec A$

- Each ratio changes to its "co-ratio": $\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \operatorname{cosec}$.

Example 4.1 Evaluate without tables: $\dfrac{\sin 25^\circ}{\cos 65^\circ} + \dfrac{\tan 23^\circ}{\cot 67^\circ}$.

SOLUTION

$\cos 65^\circ = \cos(90^\circ - 25^\circ) = \sin 25^\circ$ and $\cot 67^\circ = \cot(90^\circ - 23^\circ) = \tan 23^\circ$

$= \dfrac{\sin 25^\circ}{\sin 25^\circ} + \dfrac{\tan 23^\circ}{\tan 23^\circ}$

$= 1 + 1 = 2$

Example 4.2 Evaluate $\cos^2 20^\circ + \cos^2 70^\circ$.

SOLUTION

$\cos 70^\circ = \cos(90^\circ - 20^\circ) = \sin 20^\circ$

$\cos^2 20^\circ + \sin^2 20^\circ = 1$

Example 4.3 If $\sin 3A = \cos(A - 26^\circ)$, where $3A$ is an acute angle, find A.

SOLUTION

$\sin 3A = \cos(90^\circ - 3A)$, so $90^\circ - 3A = A - 26^\circ$

$4A = 116^\circ$

$A = 29^\circ$

5 Using the tables

- The tables of natural sines, cosines and tangents give values for angles in degrees and minutes ($1^\circ = 60'$).
- As the angle increases from 0° to 90°: sin and tan increase, so the mean difference is **added**; cos decreases, so the mean difference is **subtracted**.

6 Common mistakes

- Moving terms across the equals sign while proving. Work on one side only.
- Writing $\sin^2 A$ as $\sin A^2$.
- Cancelling $\sin A$ from a sum such as $\dfrac{\sin A + 1}{\sin A}$.
- Forgetting that $\tan(90^\circ - A)$ is $\cot A$, not $\tan A$.